INFO7375 BRANDING & AI · ASSIGNMENT 3 · DEMO

Screenshot Walkthrough

Madison Automation Agent Data Pipeline: how the n8n workflow is built, how it runs, how it handles a failing source, and what it produces.

1 Full workflow overview: a successful run

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5 Key configuration ③: the quality step

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7 Sample output ②: the final data file

Orange boxes mark key areas; numbers match the notes under each screenshot.

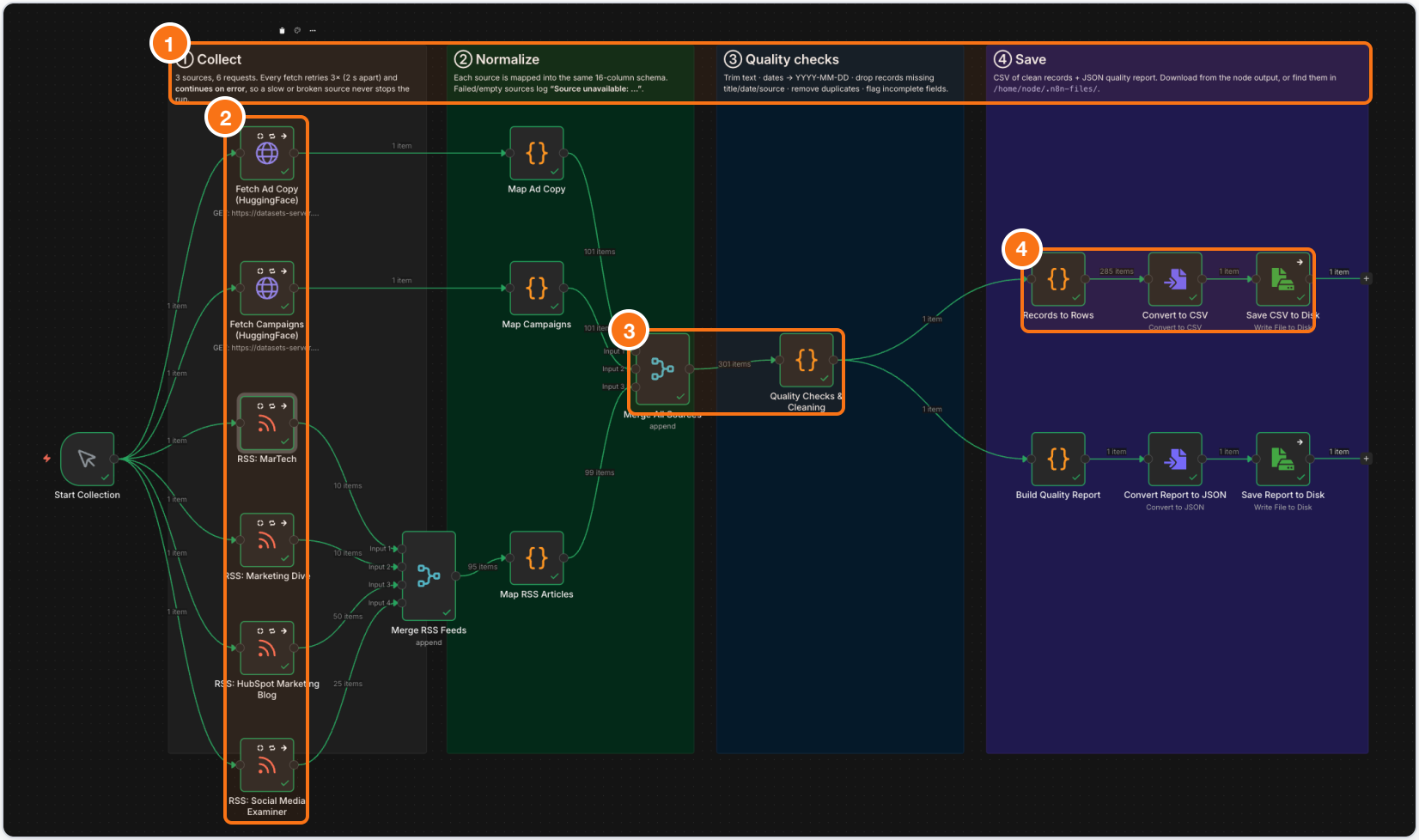
Author Mayank Bhadrasen

Workflow Bhadrasen_Mayank_A3_Workflow.json **n8n** 2.12.3 (self-hosted) **Final run** 2026-10-02

SCREENSHOT 1 OF 7

Full workflow overview: a successful run

The whole pipeline after the final run. Every node has a green check, and the item counts on each connection show the data flowing through.

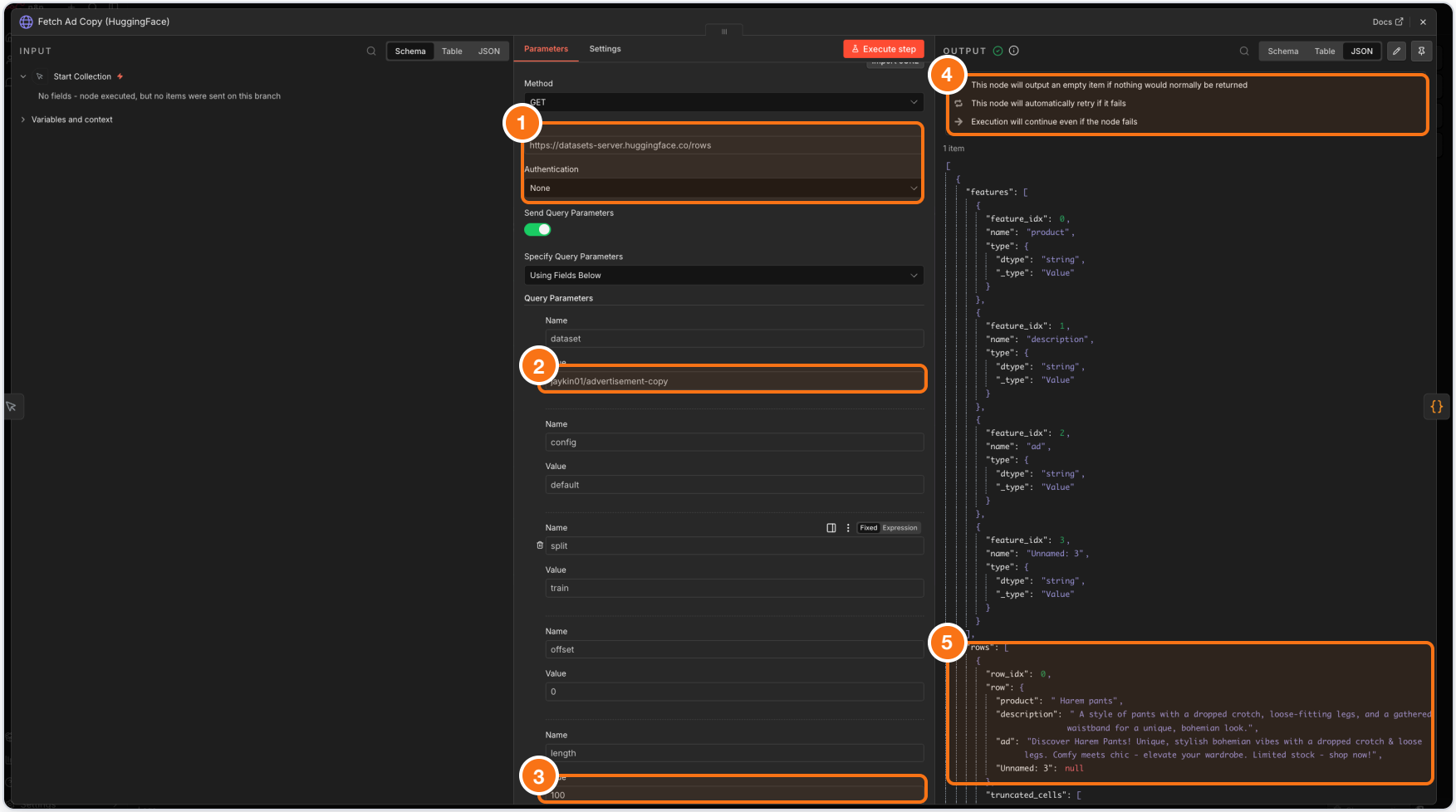


1. Four labelled sections: Collect → Normalize → Quality checks → Save. Each sticky note says what its section does.
2. Six fetch nodes run in parallel from one trigger: 2 HuggingFace datasets + 4 RSS feeds.
3. Merge All Sources combines 301 items (records + one status item per source) into the quality step.
4. 285 clean records leave the quality step and are written to CSV; the quality report runs on the branch below.

SCREENSHOT 2 OF 7

Key configuration ①: fetching a dataset with no API key

Fetch Ad Copy (HuggingFace) calls the free HuggingFace datasets-server rows API. The query parameters choose the dataset and how many rows to fetch.

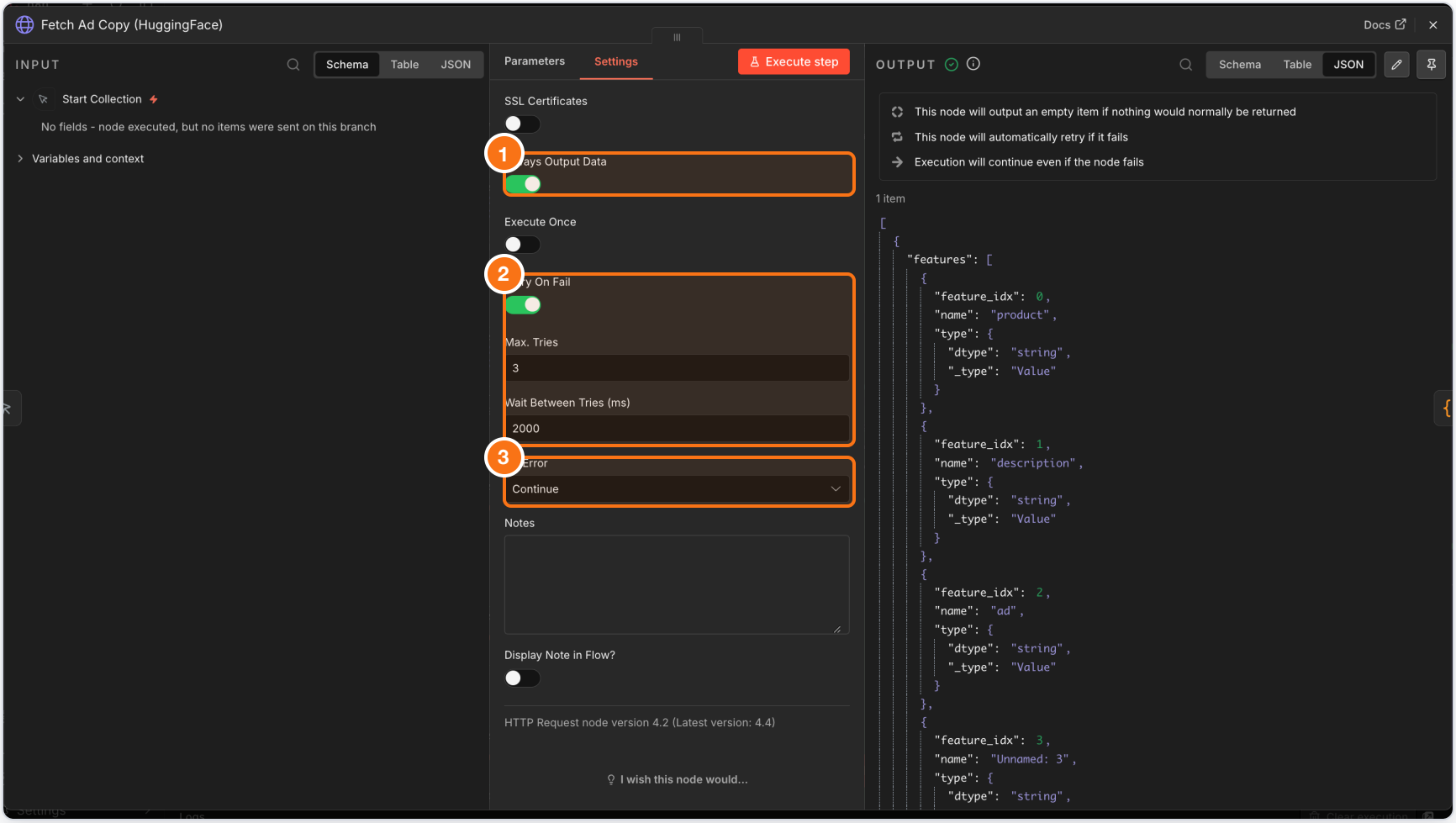


1. Public endpoint with Authentication: None. No credentials are stored in the workflow.
2. dataset: which HuggingFace dataset to read (the campaign node uses the same pattern).
3. length = 100 rows from offset 0, so every run gets the same rows.
4. n8n confirms the safety settings: always output, auto-retry, and continue on failure.
5. Live response: each row has product, description and the full ad text.

SCREENSHOT 3 OF 7

Key configuration ②: built so a slow source can't crash the run

The Settings tab of the same node. All six fetch nodes use exactly these settings.

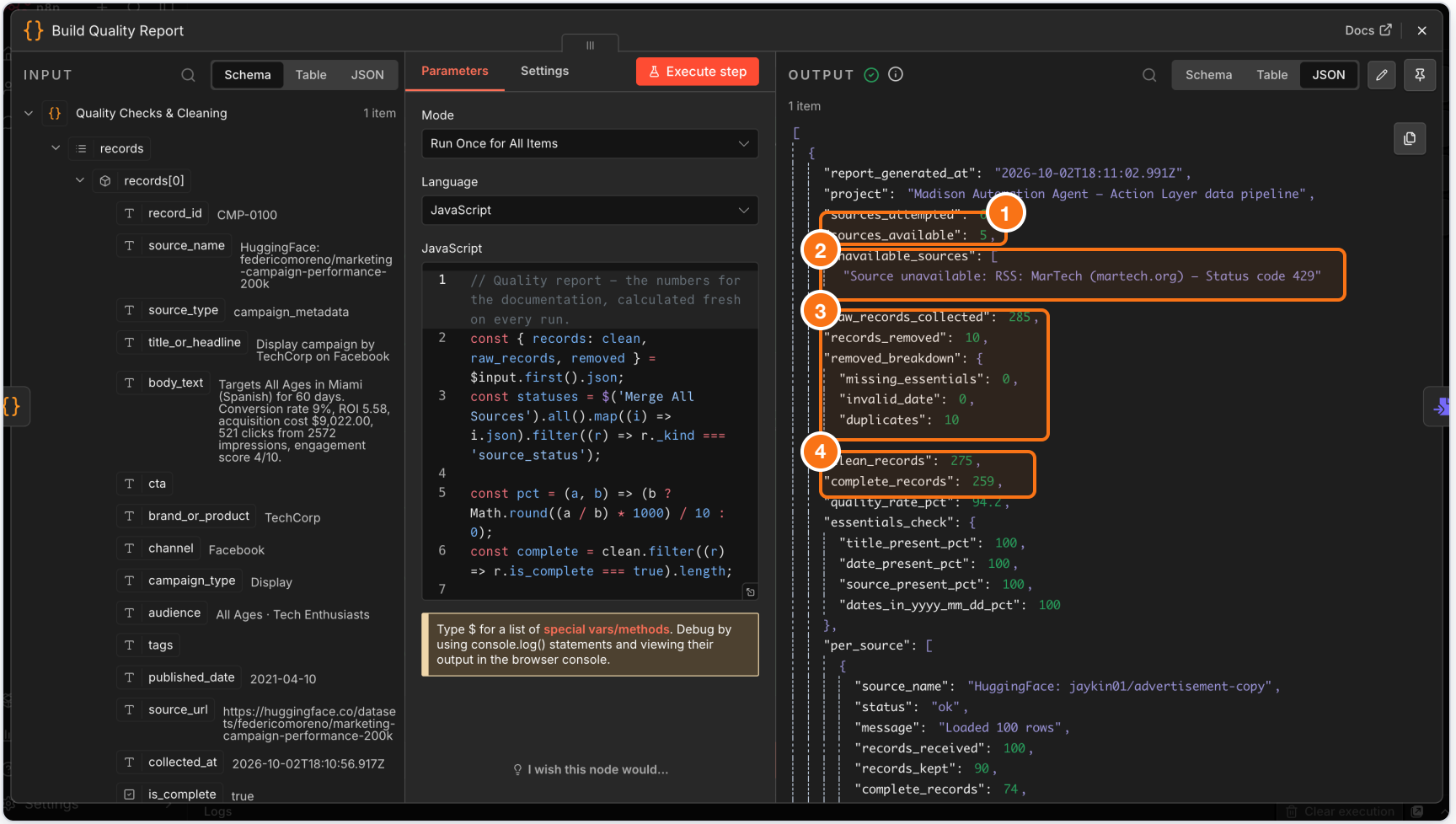


1. Always Output Data: an empty response still produces an item, so the next node can detect it and log it.
2. Retry On Fail: up to 3 tries, 2 seconds apart, for slow or flaky sources.
3. On Error → Continue: if all retries fail, the workflow moves on instead of stopping.

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Workflow running: a real source outage, handled

A test run on Oct 2 when MarTech's Cloudflare rate-limited the server (HTTP 429). Instead of crashing, the workflow logged it and finished with the other five sources.

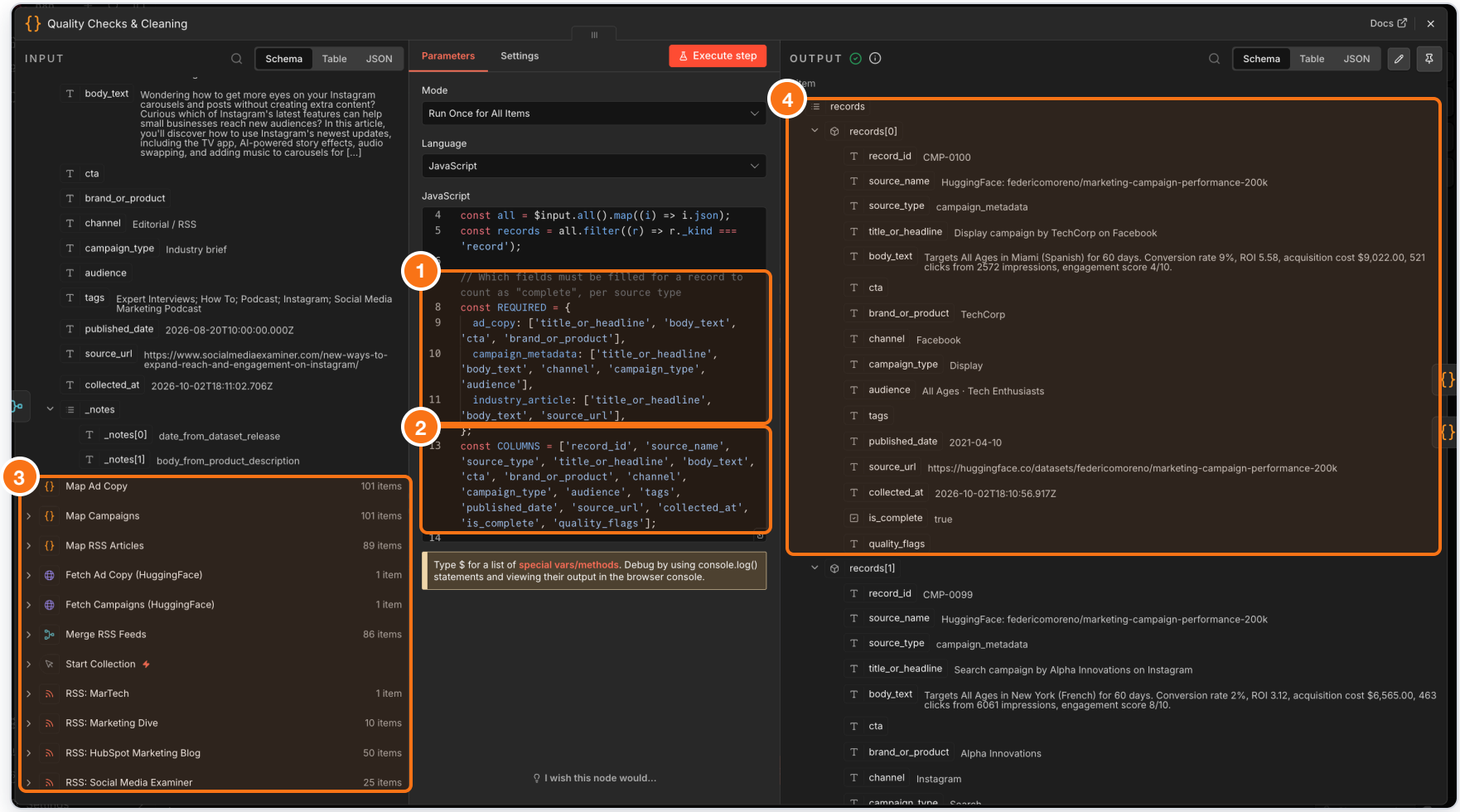


1. 6 sources attempted, 5 available on this run.
2. The built-in check logs “Source unavailable: RSS: MarTech (martech.org) – Status code 429”.
3. The run still completed: 285 records collected, 10 duplicates removed, 0 bad dates or empty records.
4. 275 clean records, 94.2% complete, without MarTech. On the final run (next screenshots) all 6 sources were available.

SCREENSHOT 5 OF 7

Key configuration ③: the quality step

Quality Checks & Cleaning runs on every record from every source: trims text, converts dates to YYYY-MM-DD, drops records missing title/date/source, removes duplicates and flags incomplete fields.

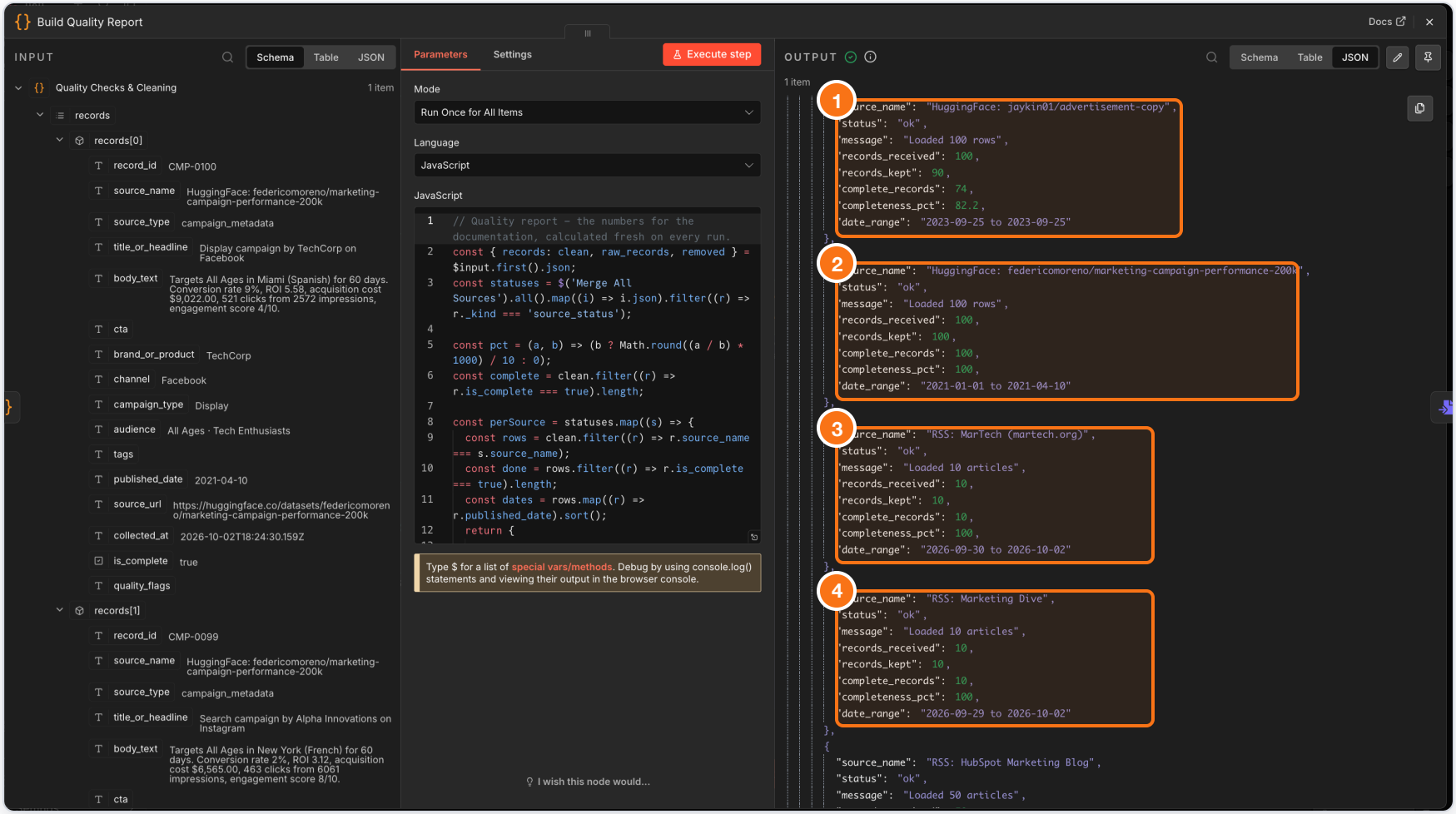


1. Required fields per source type decide is_complete. Missing ones are named in quality_flags, never filled in silently.
2. One fixed 16-column order, so the CSV is identical from row 1 to the last row.
3. Item counts from every upstream node, so you can trace each source's records through the run.
4. Output: clean, consistently shaped records ready for the CSV.

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Sample output ①: the quality report from the final run

Build Quality Report recalculates every number used in the documentation on each run. Per-source status, counts, completeness and date ranges.

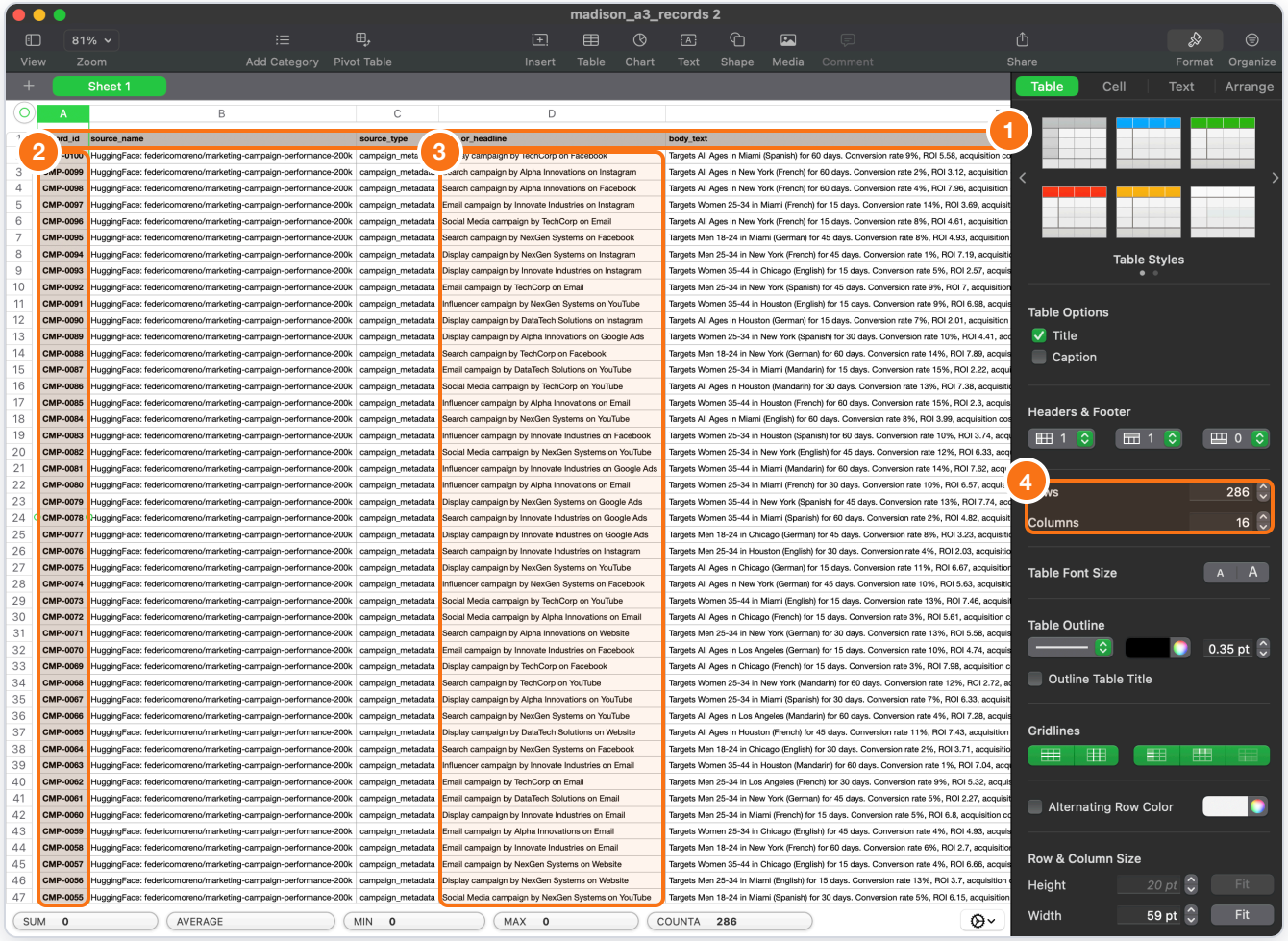


1. Ad copy: 100 received → 90 kept (the dataset repeats 10 ads with shuffled sentences) → 74 complete (82.2%).
2. Campaign metadata: 100/100 complete, dated 2021-01-01 to 2021-04-10.
3. MarTech recovered on the final run: status ok, 10 articles.
4. Marketing Dive, like every RSS feed, is reported separately with its own status and date range.

SCREENSHOT 7 OF 7

Sample output ②: the final data file

madison_a3_records.csv opened in Numbers: 285 records + header, 16 columns, one consistent schema across all sources.



1. Clear, self-explanatory column names. Each one is defined in the Data Dictionary (documentation, section 5).
2. Unique record_id; the prefix shows the source (CMP, AD, MT, MD, HS, SME).
3. Readable titles built from the raw fields, e.g. “Display campaign by TechCorp on Facebook”.
4. 286 rows (285 records + header) × 16 columns.